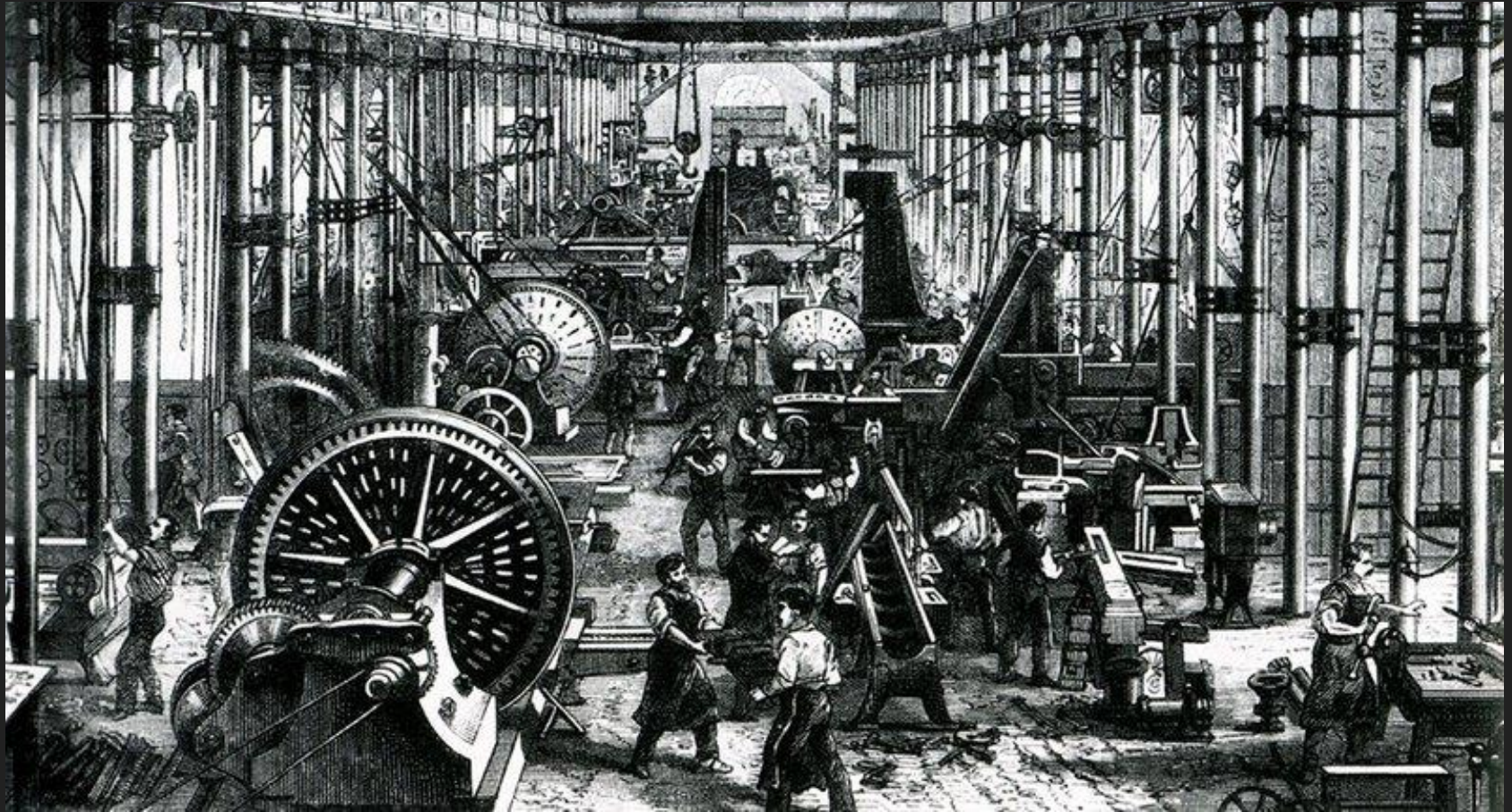


The Industrial Revolution



Lesson 1 Objectives

Identify specific causes of the Industrial Revolution

Analyze the relationship between the Agricultural Revolution and urbanization and the impacts of that relationship

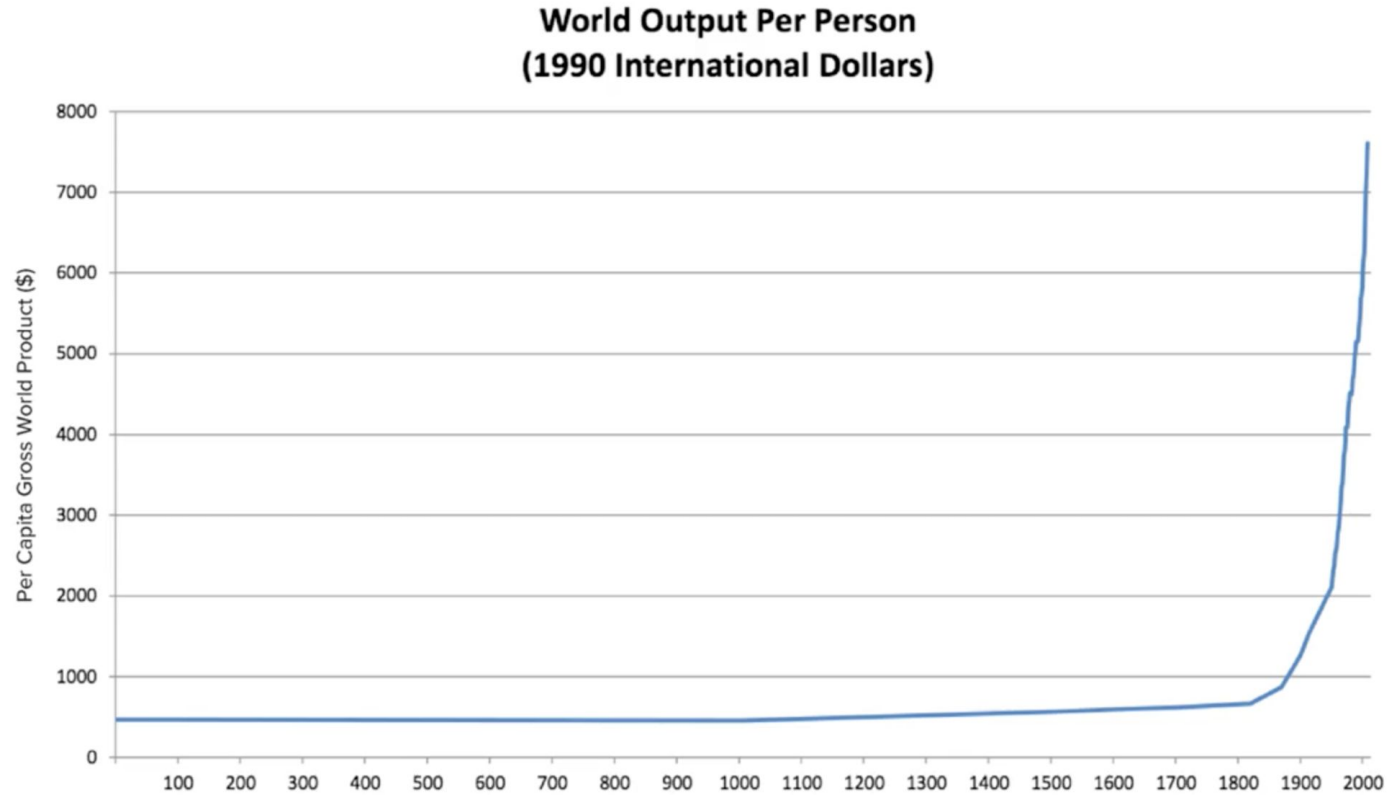
Identify reasons for England's early success in the Industrial Revolution

Background Information

1st Industrial Revolution: ~1750 - ~1840

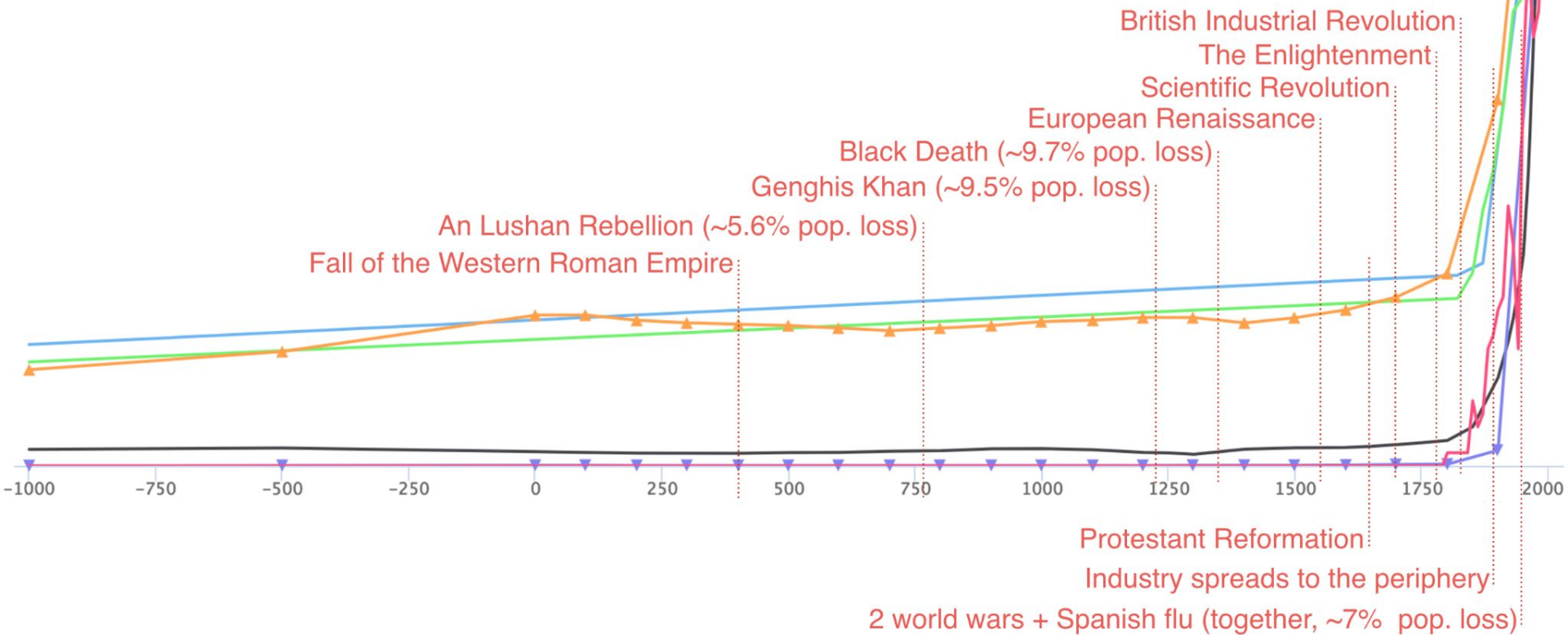
2nd Industrial Revolution - ~1850s - 1914

- Slow and uneven
- Agriculture to industry
- Wielding industry in new ways



Our World in Data (based on Maddison Database and World Bank)

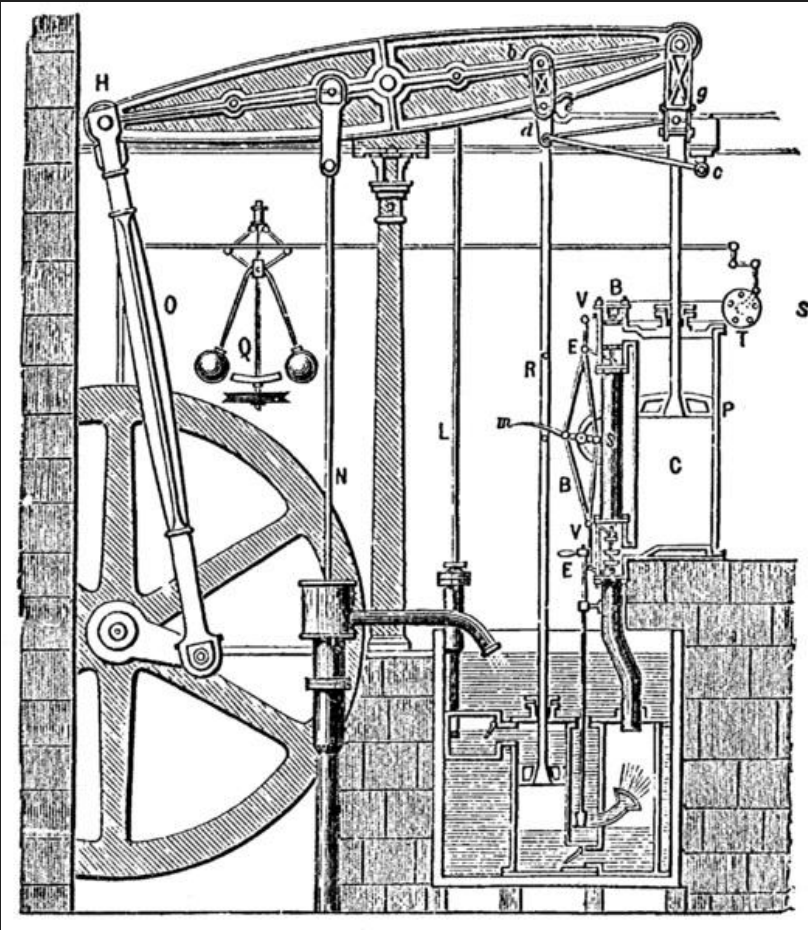
- Life expectancy
- GDP/cap
- % not living in extreme poverty
- Energy capture (kcal/cap/day)
- War-making capacity
- % living in a democracy



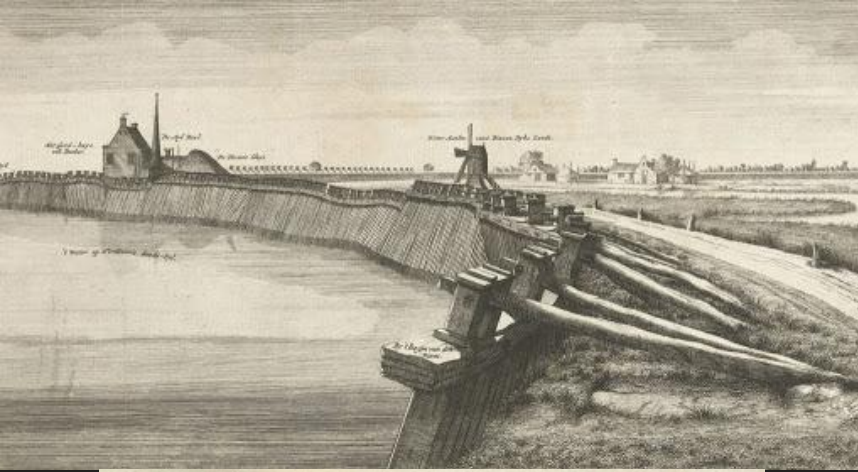
What caused the Industrial Revolution?



Basic Causes

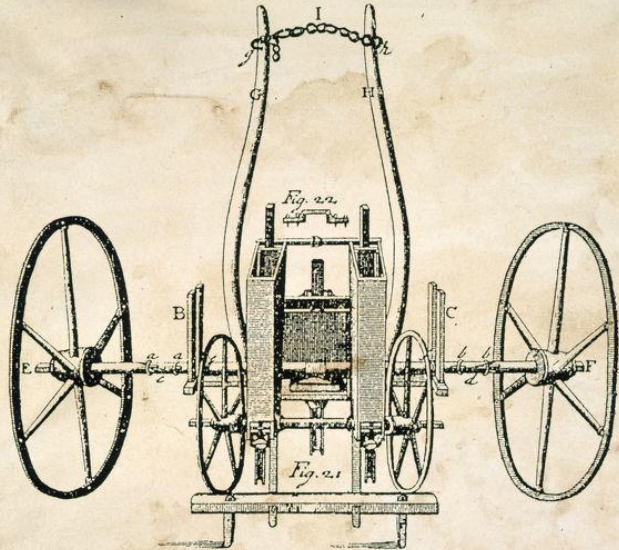


- 2nd Agricultural Revolution
- Coal, Steam, and the Energy Revolution
- Various simultaneous factors



2nd Agricultural Revolution

- Quality and quantity of farm products
- Led by Dutch farmers
 - Dikes, larger fields, and fertilization
- British mix soils and try crop rotation
 - Growing turnips and clover
 - Jethro Tull and effective planting



JETHRO TULL'S WHEAT DRILL

2nd Agricultural Revolution

- Practice of enclosures
 - Large-scale farming
 - Cheaper, more efficient
 - Mass unemployment
 - Must find jobs in the cities
- People are healthier and live longer
 - Less famine, disease, etc.



Coal, Steam, and the Energy Revolution

- James Watt advances Thomas Newcomen's model
- Coal is necessary
- Coal can be used to smelt iron more effectively
 - Pull iron away from its ore
 - Cheaper and more effective metal

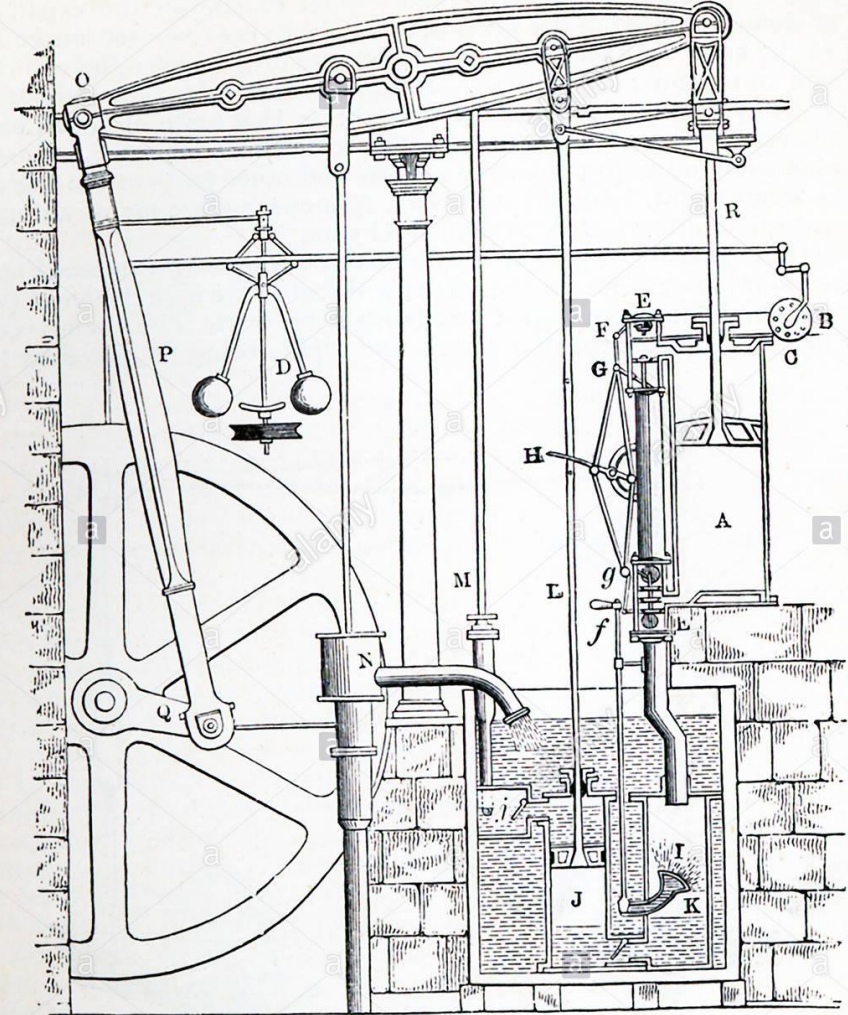


FIG. 3.—Watt's Double-action Steam Engine.

Other Necessities

- Natural Resources
- Capital
- Labor
- Entrepreneurship
- Demand



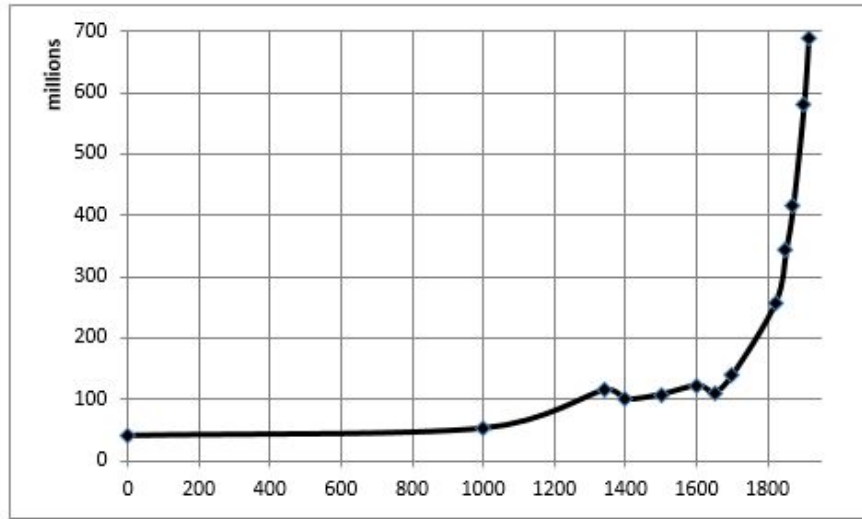


Fig. 2. Population dynamics of Greater Europe, millions of people, 1–1913 CE

Sources: Maddison 2010; data point for 1650 is calculated taking into account the estimates of the 17th-century depopulation presented by Geoffrey Parker (Parker 2013).

- Natural Resources
 - Coal and iron readily available
 - Island empire with many ports and connections to the Americas and slave production
- Capital, Labor, and Demand
 - Cyclical nature of profits
 - Agriculture leads to births leads to demand
 - Businesses are willing to risk money for potential gains

- Entrepreneurship

- Inventors and businesses are flourishing
- Government is stable and allows for economic growth
 - Strong navy helps with trade

- England has all of the factors we discussed



A Transportation Revolution

Transportation Revolution

- Turnpikes, canals, railroads, and bridges
 - Cheaper and easier transportation leads to cheaper goods
 - Leads to more demand

Figure 7: Railways and local population growth from 1851 to 1881

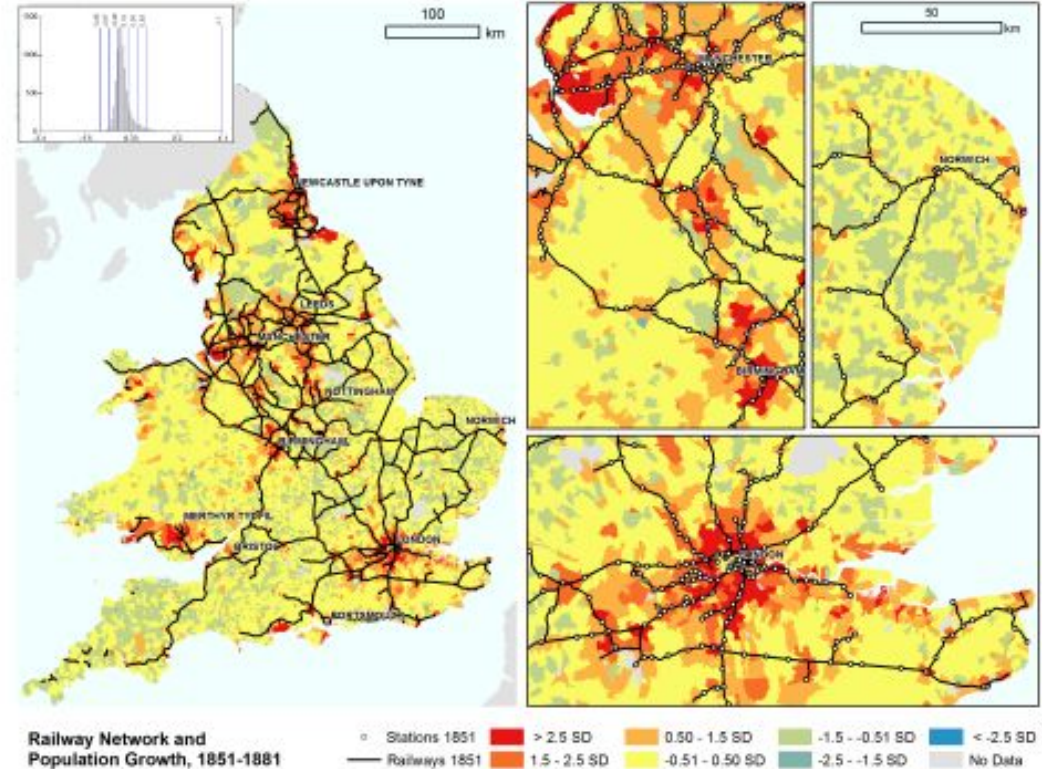
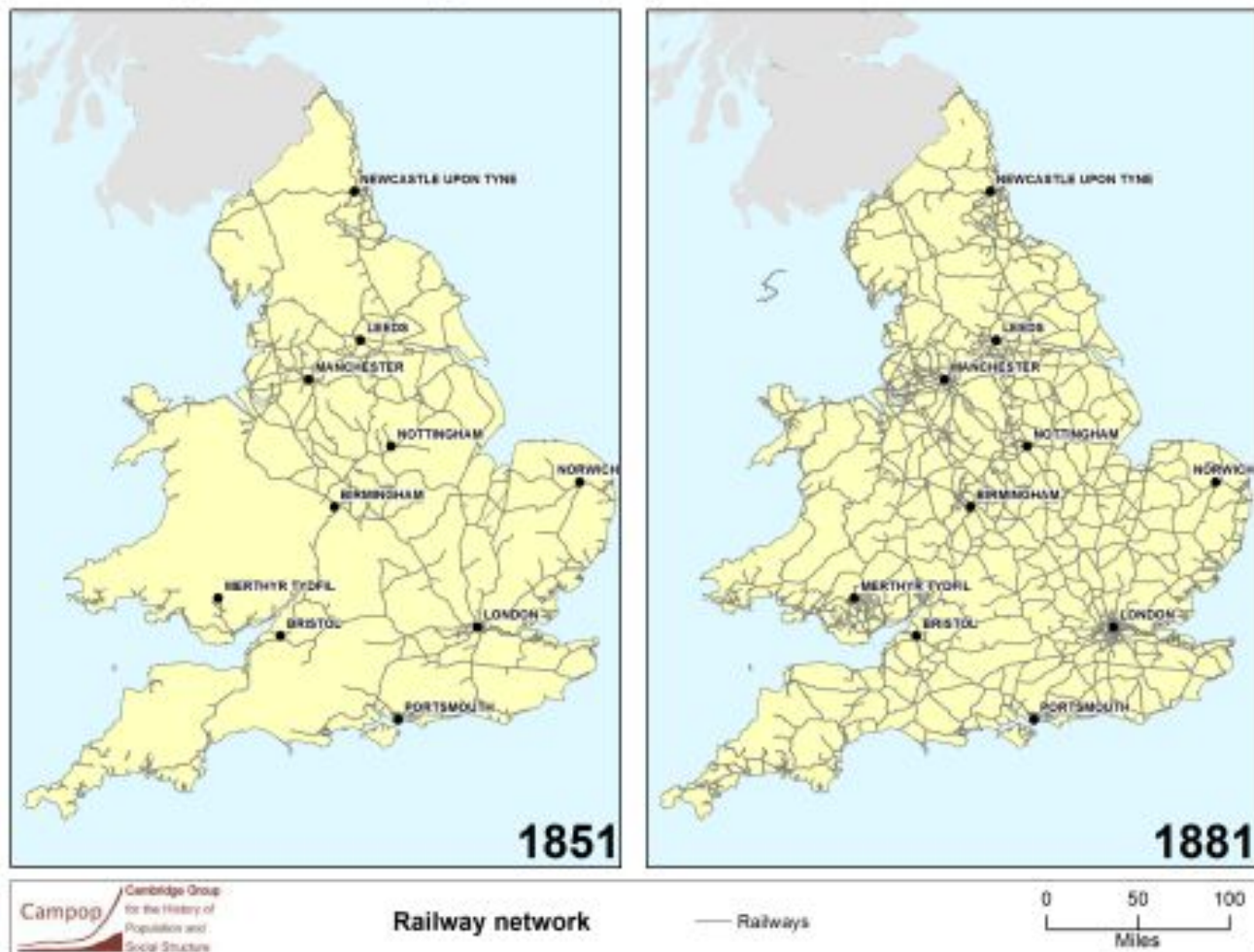


Figure 3: Railway lines and stations in 1851 and 1881



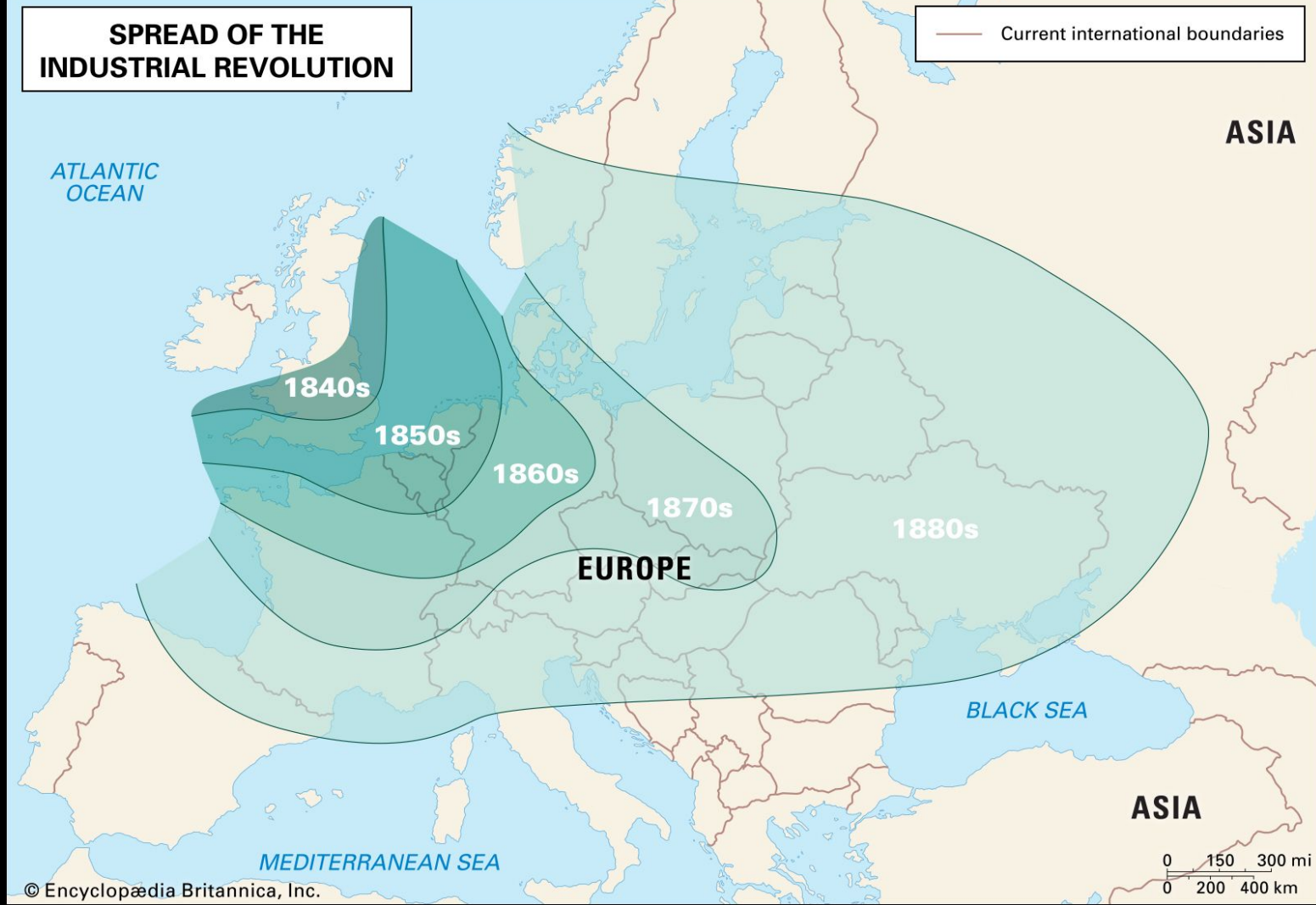
Spread of Industrialization

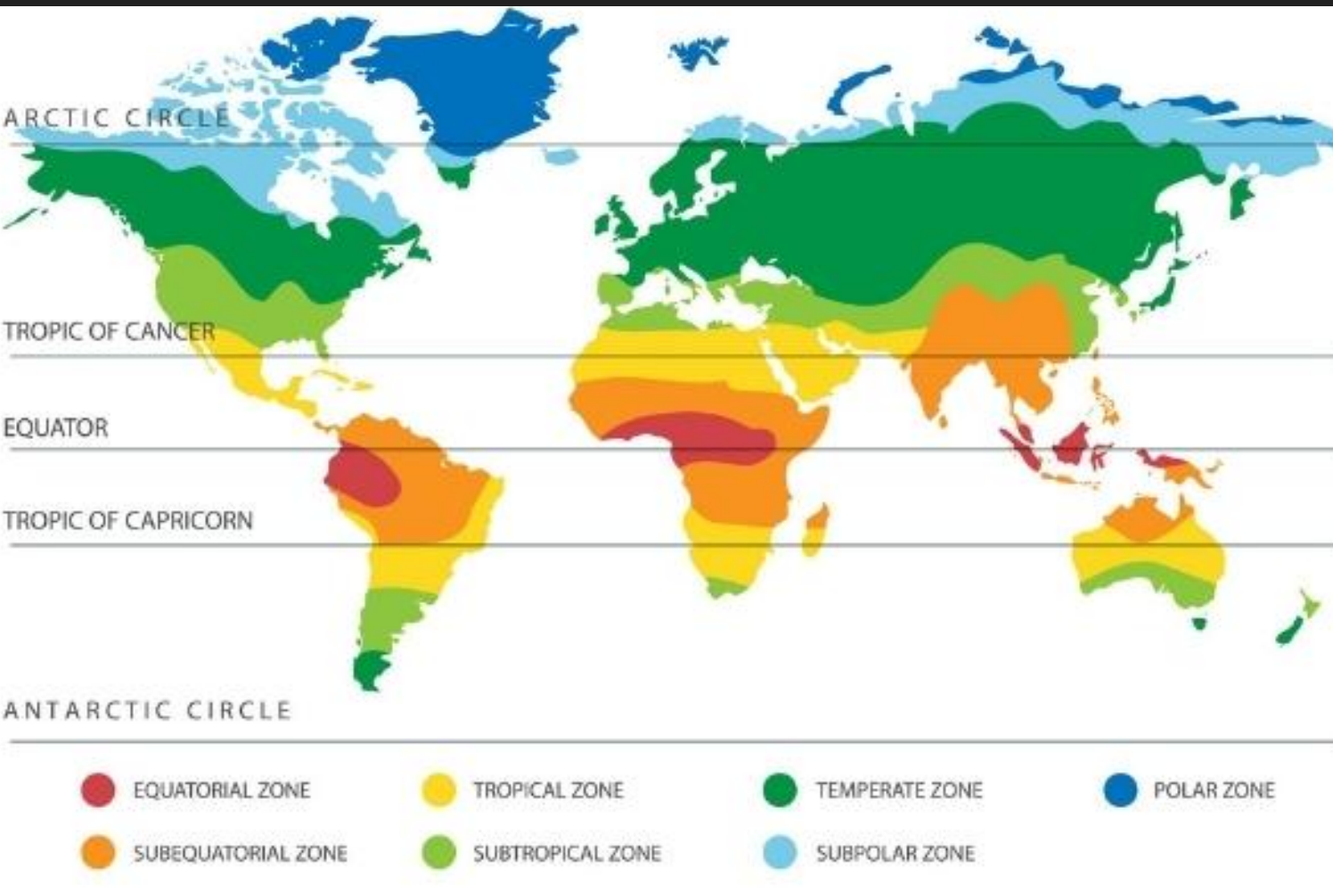
- England and most of Western Europe will industrialize very quickly
 - US included as well
- Industry spreads unevenly globally
 - Lack of resources needed
 - Lack of stability in governments
- Social, economic, and political changes



SPREAD OF THE INDUSTRIAL REVOLUTION

— Current international boundaries





Agricultural and economic changes also occurred in others parts of the world including China and Japan

Lesson 1 Objectives

Identify specific causes of the Industrial Revolution

Analyze the relationship between the Agricultural Revolution and urbanization and the impacts of that relationship

Identify reasons for England's early success in the Industrial Revolution

Lesson 2 Objectives

Identify changes in the population landscape of industrial societies

Compare and contrast the different social thinkers of the time and their beliefs about population, society, and government

Assess the ways in which the industrial revolution brought positive changes, but also had negative consequences

Social Impacts of the Industrial Revolution

Changes to Society

- Rapid Urbanization
 - Small towns to major cities
 - Growth around coal mines and rail hubs
- New Social Hierarchy
 - Bourgeoisie, middle class inventors, entrepreneurs, merchants, artisans, etc.



Britain (mid 1700s)



Britain (late 1800s)

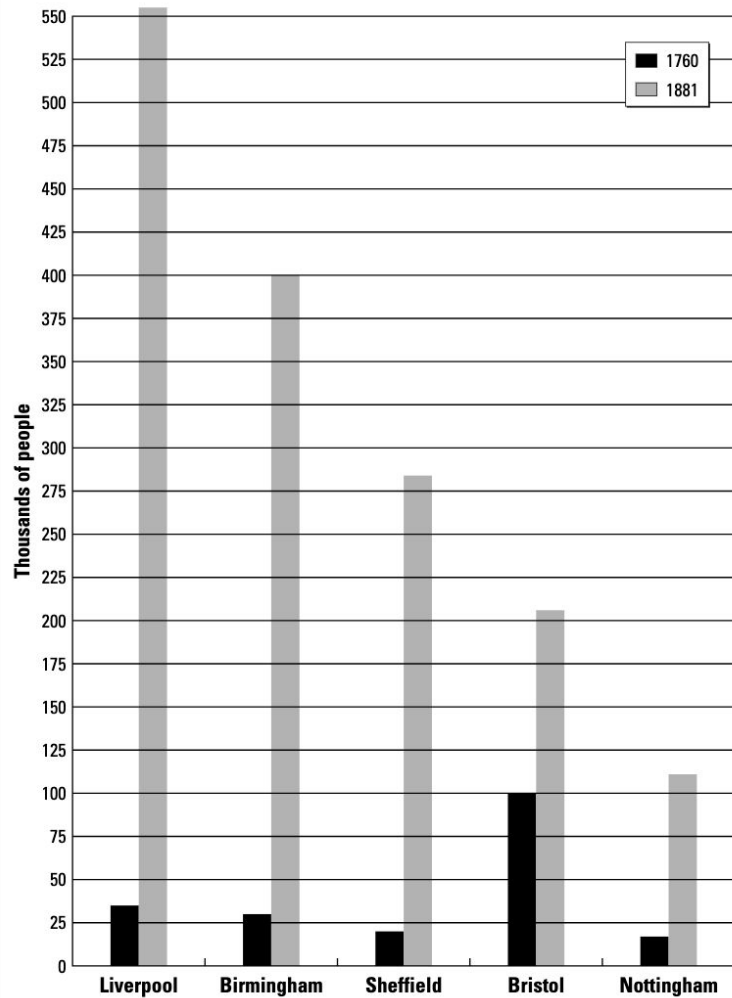


Population of Selected British Cities 1801–1891

Town	1801	1861	1891
Birmingham	74,000	296,000	523,000
Leeds	53,000	207,000	429,000
Liverpool	80,000	444,000	704,000
Manchester	90,000	339,000	645,000

Source: B.R. Mitchell, *International Historical Statistics: Europe, 1750–1988*, Stockton Press, Third Edition
(adapted)

Growth of British Cities 1760–1881



SOCIAL CLASS CHANGES IN ENGLAND

Class was based on social rank as well as income. Manners, clothing, background, and way of life all distinguished the various classes. For example, a teacher might make less money than a farmer, but socially the teacher would be in a higher class.

17TH-CENTURY ENGLAND

Titled nobility
such as dukes
and earls

Merchants

Laborers
and farmers

19TH-CENTURY ENGLAND

UPPER CLASS

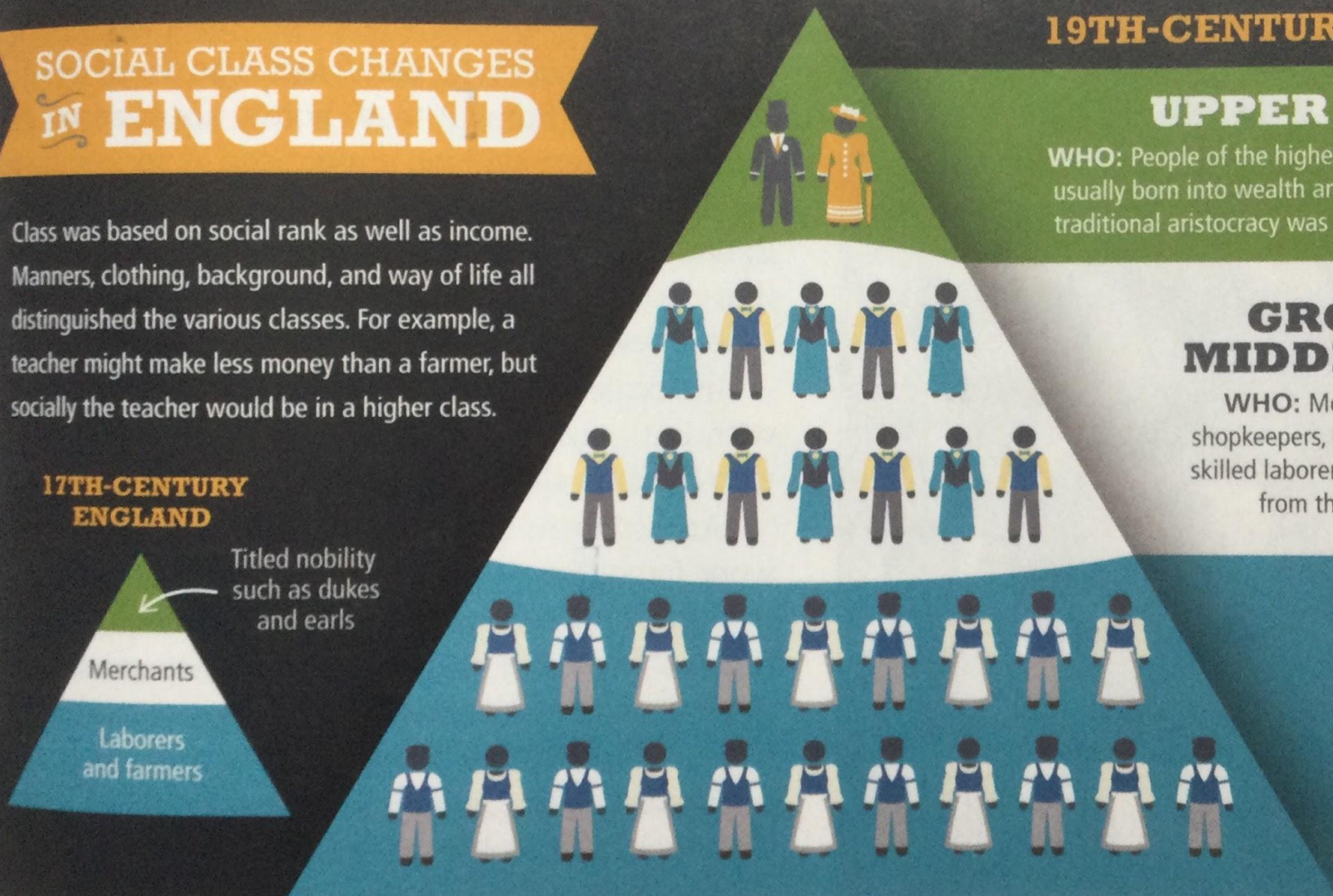
WHO: People of the highest level of society who are usually born into wealth and are the most educated; traditional aristocracy was based on land ownership

GROWING MIDDLE CLASS

WHO: Merchants, inventors, shopkeepers, bankers, teachers, and skilled laborers, some of whom rose from the working class

WORKING CLASS

WHO: Laborers such as agricultural workers, factory workers, and servants



Changes to Society

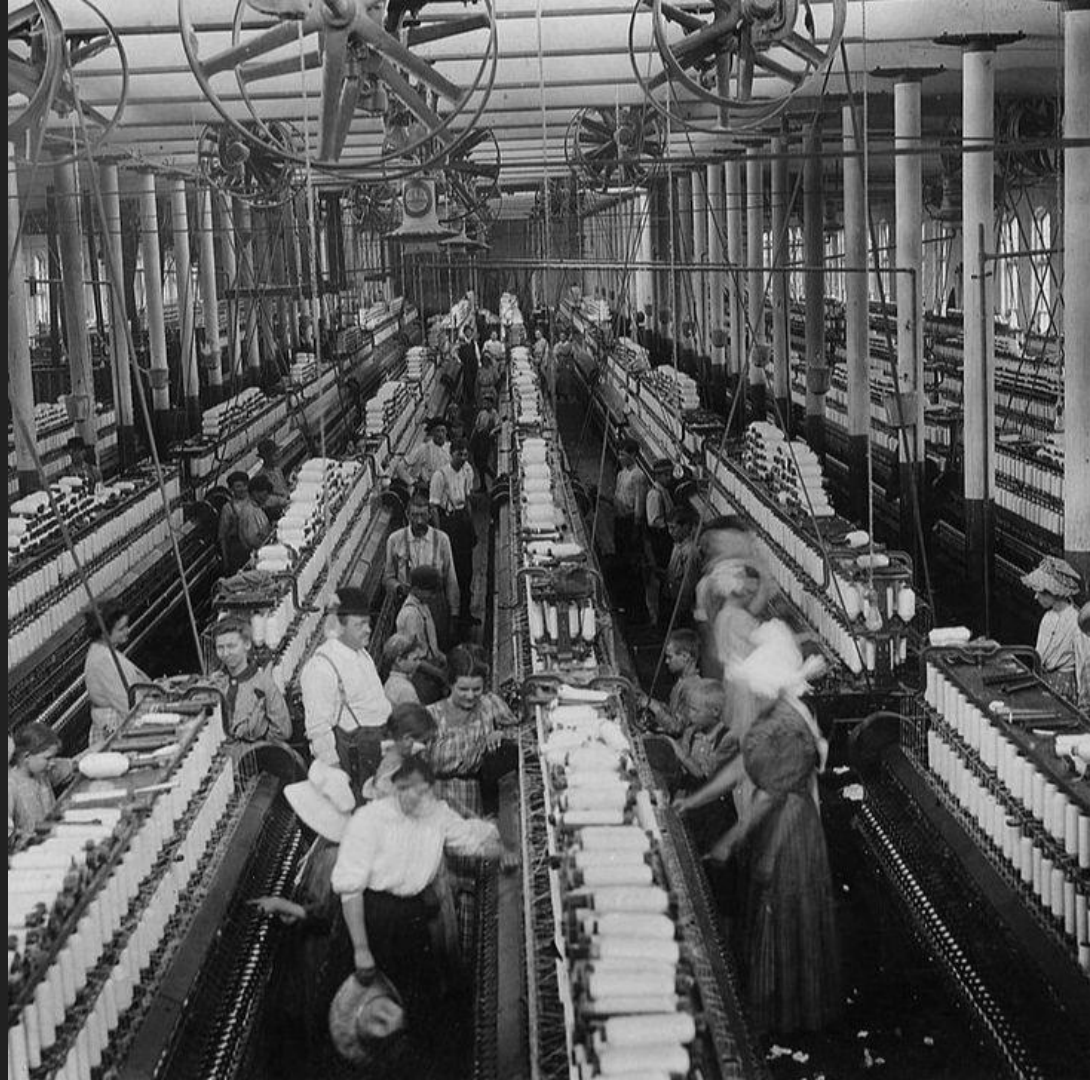


- Workers have minimal rights
 - No labor unions
- Poorest live in tenements
- Religious institutions offer support to the poor

Industrial Working Conditions

The Textile Industry

- Originally used the “putting-out” system
 - Raw materials to independent workers
 - Finishing touches put by experts
- Inventions drastically speed up production
 - Flying shuttle, spinning jenny, cotton gin
 - Factory system with generic pieces





Industrial Working Conditions

- Long hours, dangerous
 - Minimal Pay
- Women and children preferred to men in many cases
 - Mines, thread factories
- Exposed by the Sadler Report









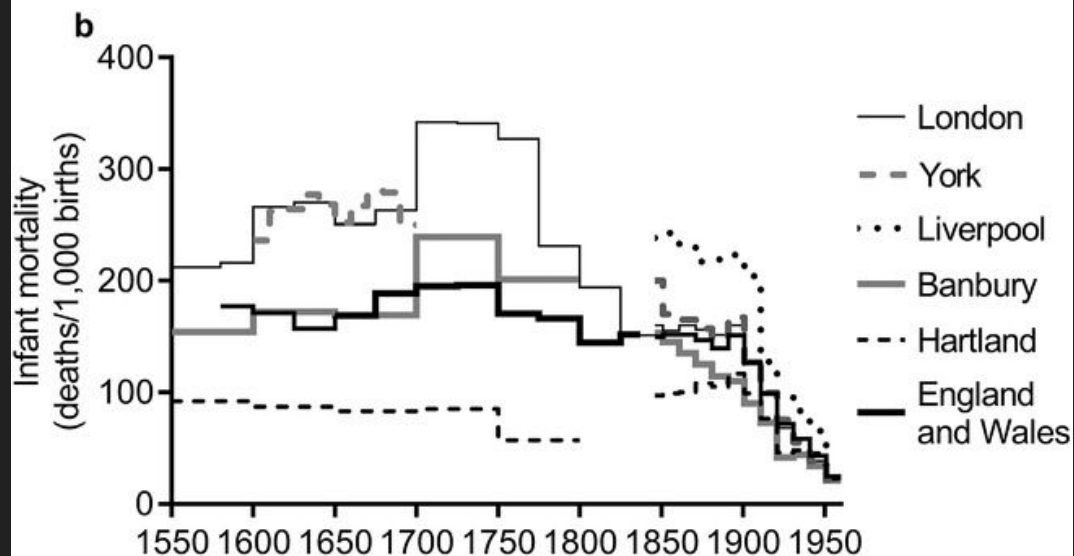
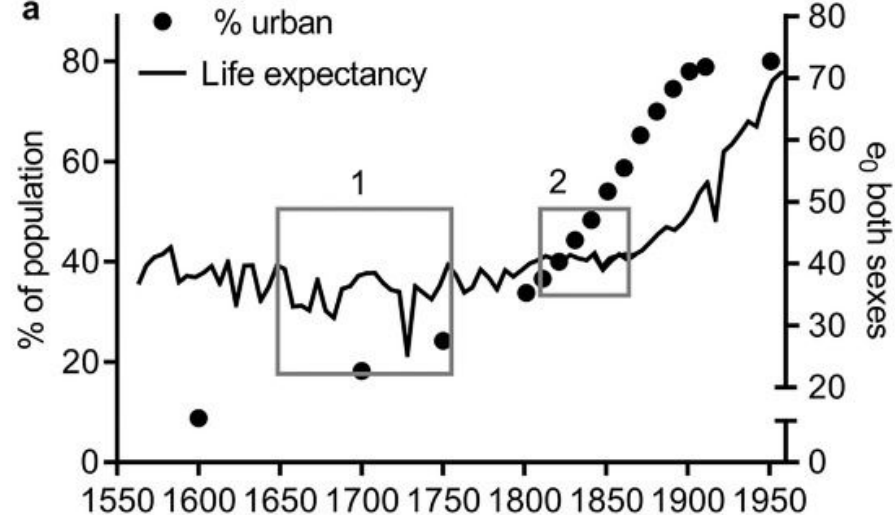
THE REVOLTING CASE OF BETTY HARRIS.



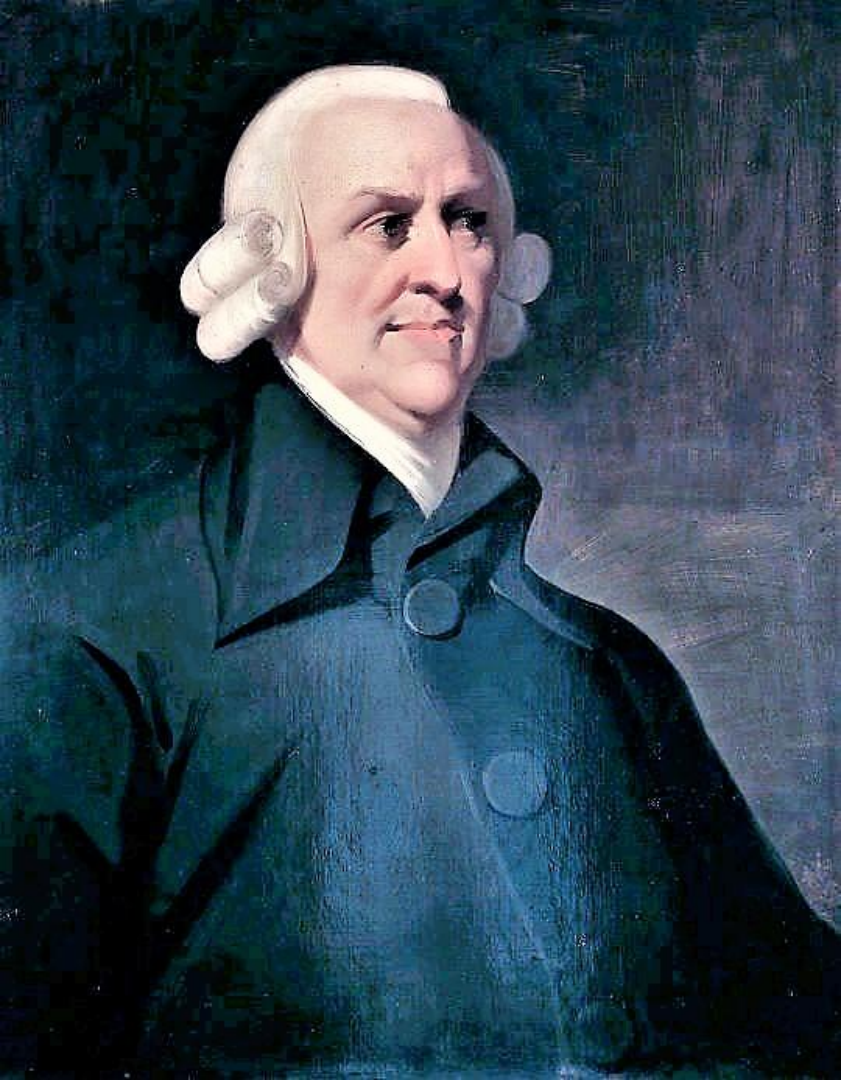
Benefits of the Industrial Revolution

- Improved standards of living
 - Cheaper, more readily available goods
- Chances of “rags to riches” story
 - More open to change if it means making more money
 - Social Mobility





Emergence of Social Thinkers



Adam Smith

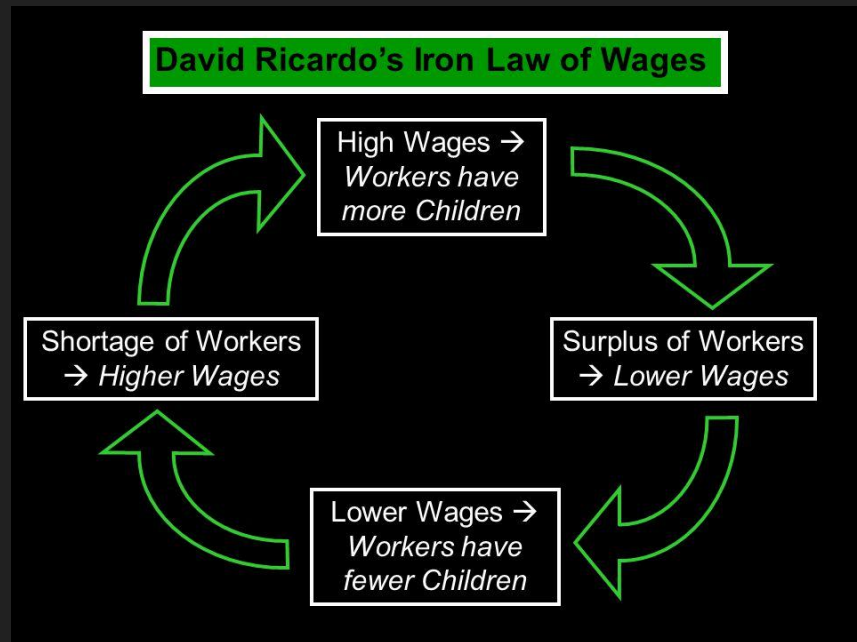
- Laissez-Faire Economics
 - Free Market, or unregulated exchange of goods and services
- Government plays a limited role in financial success

Thomas Malthus and David Ricardo



Malthus, Ricardo, and Population Sizes

- Thomas Malthus
 - Growth needed to be checked by “natural forces” so as not to outpace the food supply
 - Urged families to have fewer children
- Ricardo believed in the “Iron Law” and that the poor were having too many children
- Both opposed government help for the poor



Utilitarians and Utopians



UTILITARIANISM

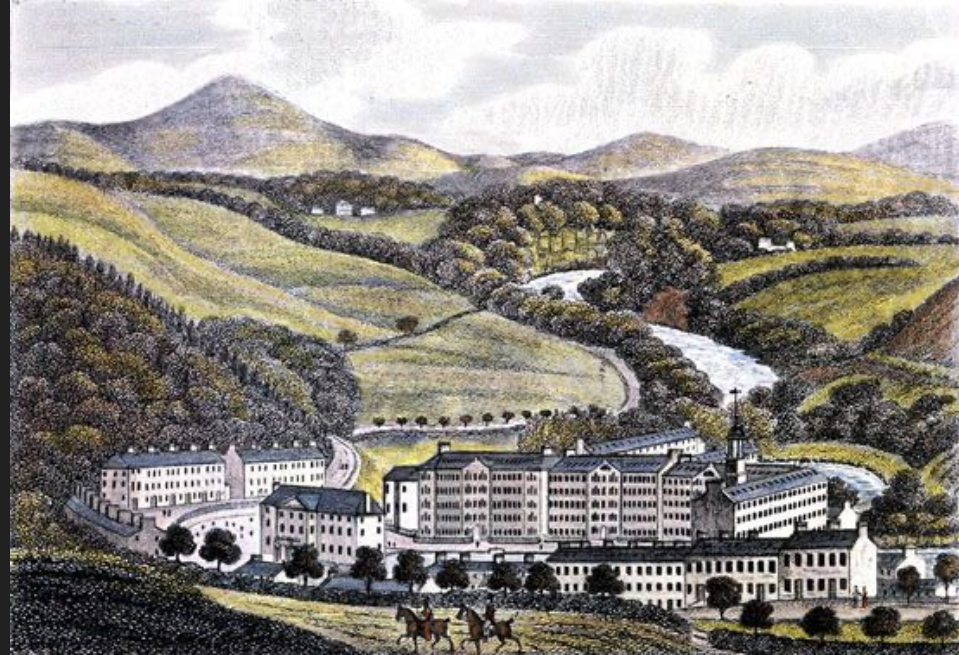
“Nature has placed mankind under the governance of two sovereign masters, pain and pleasure. It is for them alone to point out what we ought to do, as well as to determine what we shall do.”

- Jeremy Bentham

- Jeremy Bentham's Utilitarianism
 - Gov. should act for greatest happiness for greatest number of people
 - Individual freedom was essential
- John Stuart Mill
 - Gov should act as support for working classes
 - Owners should only better themselves if they don't hurt workers

Utopians

- Robert Owen
 - Introduced a version of socialism
 - People as a whole own the means of production (farms, factories, railways, etc)
 - New Lanark, Scotland -- The First Utopia



Karl Marx and Friedrich Engels





CRASH COURSE SOCIOLOGY

6

KARL MARX & CONFLICT THEORY

	Socialism	Fascism	Capitalism	Communism
OWNERSHIP	Individuals may own property and small businesses. State ownership of essential services (electricity, water, internet)	Individuals may own property and small businesses. State ownership of essential services through dictator	Individuals have right to own property. Little to no government ownership of services or production	Entire elimination of property rights – no individual land ownership. No economic incentives to individually own
EQUALITY	Redistribution of income to create equal classes. Total freedom of religion	Nationalistic religion. Unequal redistribution of wealth to reinforce a rigid class structure	Freedom of movement and a free economy gives people equal opportunity to pursue life	All are equal under the eyes of the government except the ruling class, which is in charge of the redistribution of wealth
PRICES	Redistribution of wealth through government oversight. Price control on staple goods while much of economy runs unfettered	Prices set around class structure and are influenced by supply and demand.	Prices are set by consumer/corporation through supply and demand. Competition brings with it lower prices and better products.	Prices defined by government who respond and control supply and demand to artificially control pricing.
TAXES	High taxes focused on social services or welfare state	High taxes focused on raising military funds for national defense	Limited taxes to match limited government	High, strict tax rates focused on redistributing wealth to central power
HEALTHCARE	Free and accessible to all. Operated by the federal government	Accessible and free to all that meet requirements (race, nationality, citizenship are usually factors)	Determined by market – little to no government regulation.	Accessible to all but distributed unequally (less accessible to poor). Operated by the federal government
NOTABLE FIGURES	César Chavez, Albert Einstein, Friedrich Engels	Benito Mussolini, Adolf Hitler, Nikola Mandic	Adam Smith, Ronald Reagan, George Washington	Karl Marx, Fidel Castro, Joseph Stalin, Vladimir Lenin
GENERAL PHILOSOPHY	Internationalistic with idealized future based on redistribution of wealth to fund and support social programs	Nationalistic agenda usually centered around one powerful individual with an idealized past that encourages individualism	The free market will determine prices and competition will drive improvements	Capitalism has failed to help the working class because it only incentivizes individualism and therefore must be torn through force

Lesson 2 Objectives

Identify changes in the population landscape of industrial societies

Compare and contrast the different social thinkers of the time and their beliefs about population, society, and government

Assess the ways in which the industrial revolution brought positive changes, but also had negative consequences

Lesson 3 Objectives

Identify reasons for the 2nd Industrial Revolution occurring

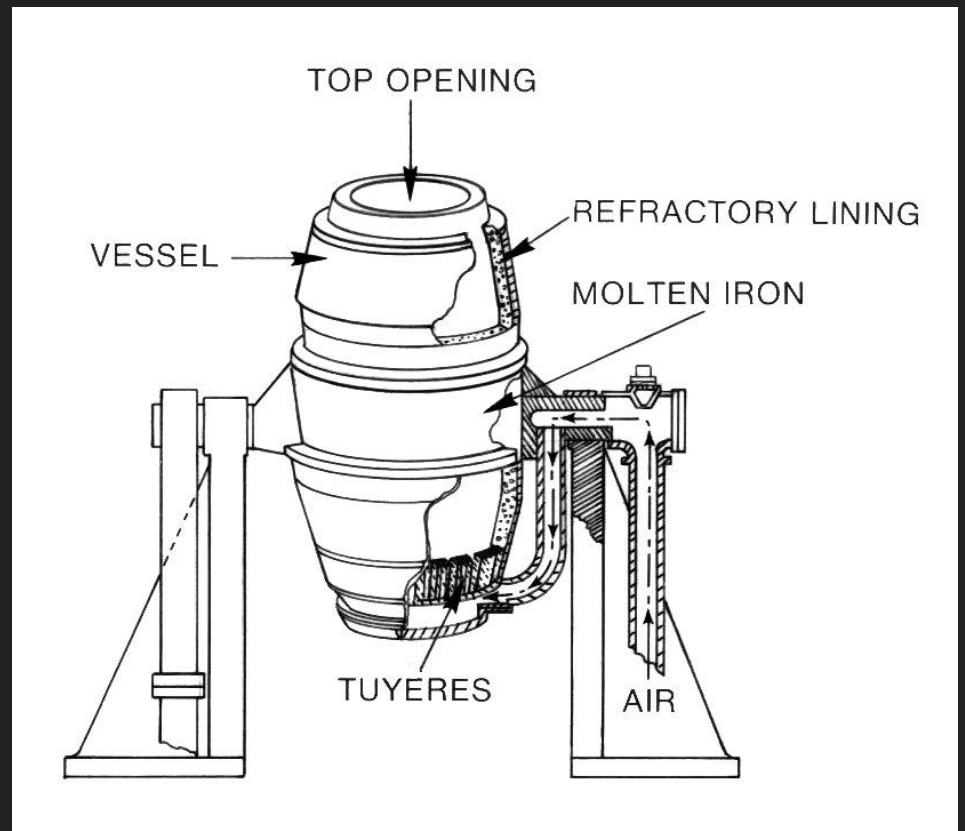
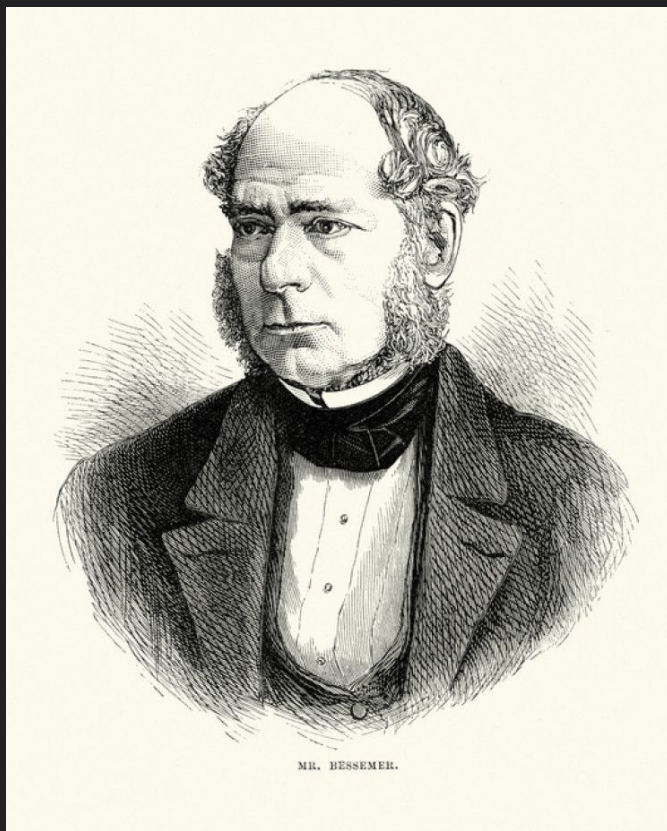
List examples of technological improvements during the
2nd Industrial Revolution

The Second Industrial Revolution

How is this different?

- Science, technology, and industry will all combine to spur economic growth
 - Jobs centered around these fields
- Improvement upon already existing methods

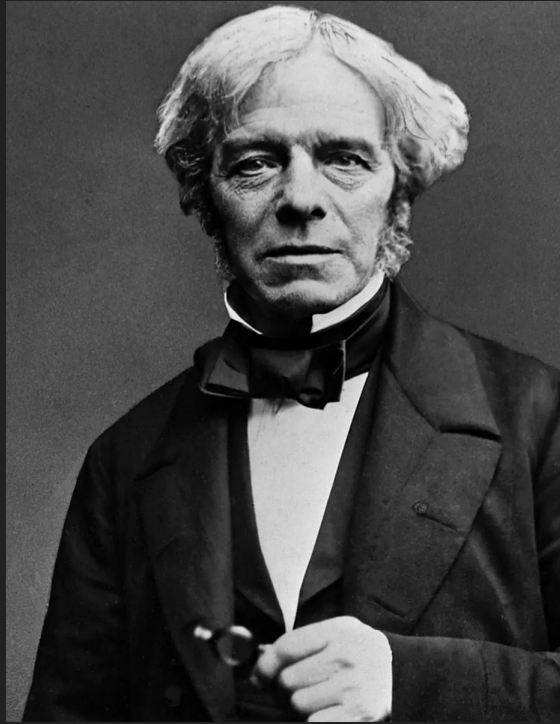




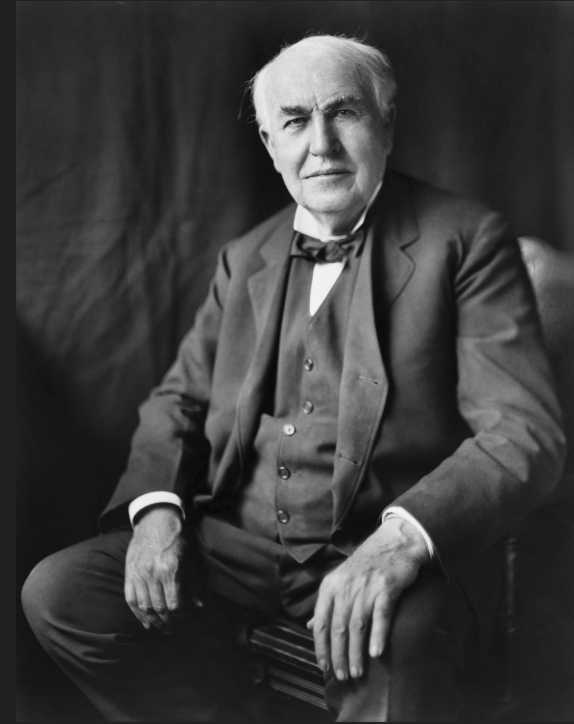
Henry Bessemer and the Creation of Steel - a more durable and effective metal



Dynamite for mines



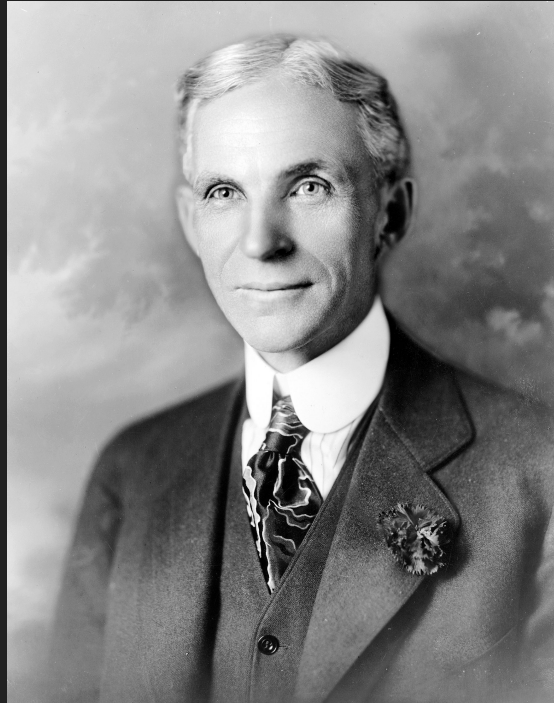
Dynamo Machine for electric motor



Light bulb

Inventions for Efficiency and Safety
Alfred Nobel, Michael Faraday, and Thomas Edison

Interchangeable Parts and the Assembly Line



Advances in Transportation and Communication



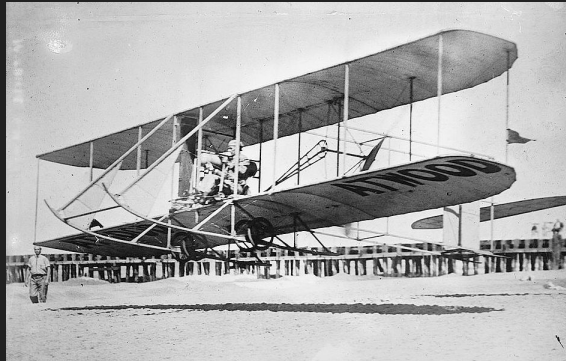
Nikolaus Otto and
the combustion
engine



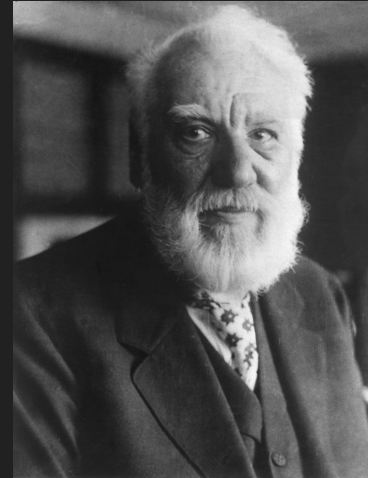
Guglielmo
Marconi and the
Radio (LEFT)

Samuel Morse's
telegraph
alphabet (RIGHT)

Alexander
Graham Bell and
the telephone
(BELOW)



Wilbur and Orville
Wright and the first
airplane in NC



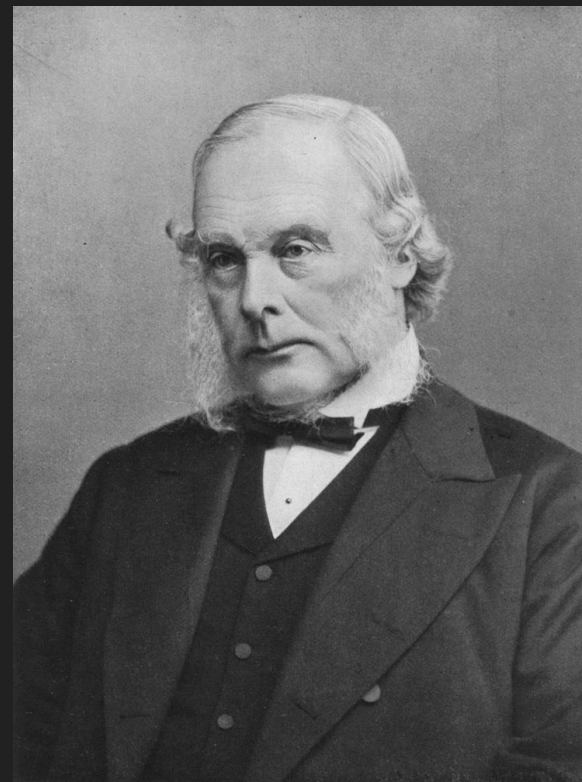
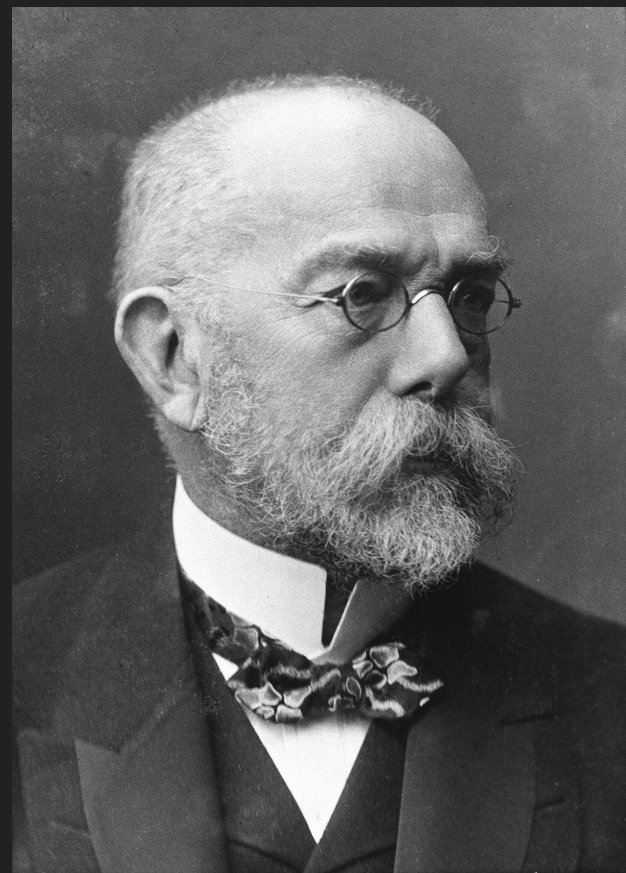
INTERNATIONAL MORSE CODE

1. A dash is equal to three dots.
2. The space between parts of the same letter is equal to one dot.
3. The space between two letters is equal to three dots.
4. The space between two words is equal to five dots.

A	U
B	V
C	W
D	X
E	Y
F	Z
G	
H	
I	
J	
K	1
L	2
M	3
N	4
O	5
P	6
Q	7
R	8
S	9
T	0



Louis Pasteur and Germ Theory



Robert Koch (TB), Florence Nightingale (Hygiene), and Joseph Lister (Antiseptics)

Lesson 3 Objectives

Identify reasons for the 2nd Industrial Revolution occurring

List examples of technological improvements during the
2nd Industrial Revolution

Lesson 4 Objectives

Identify ways in which the Industrial Age reshaped the social and economic structures of cities

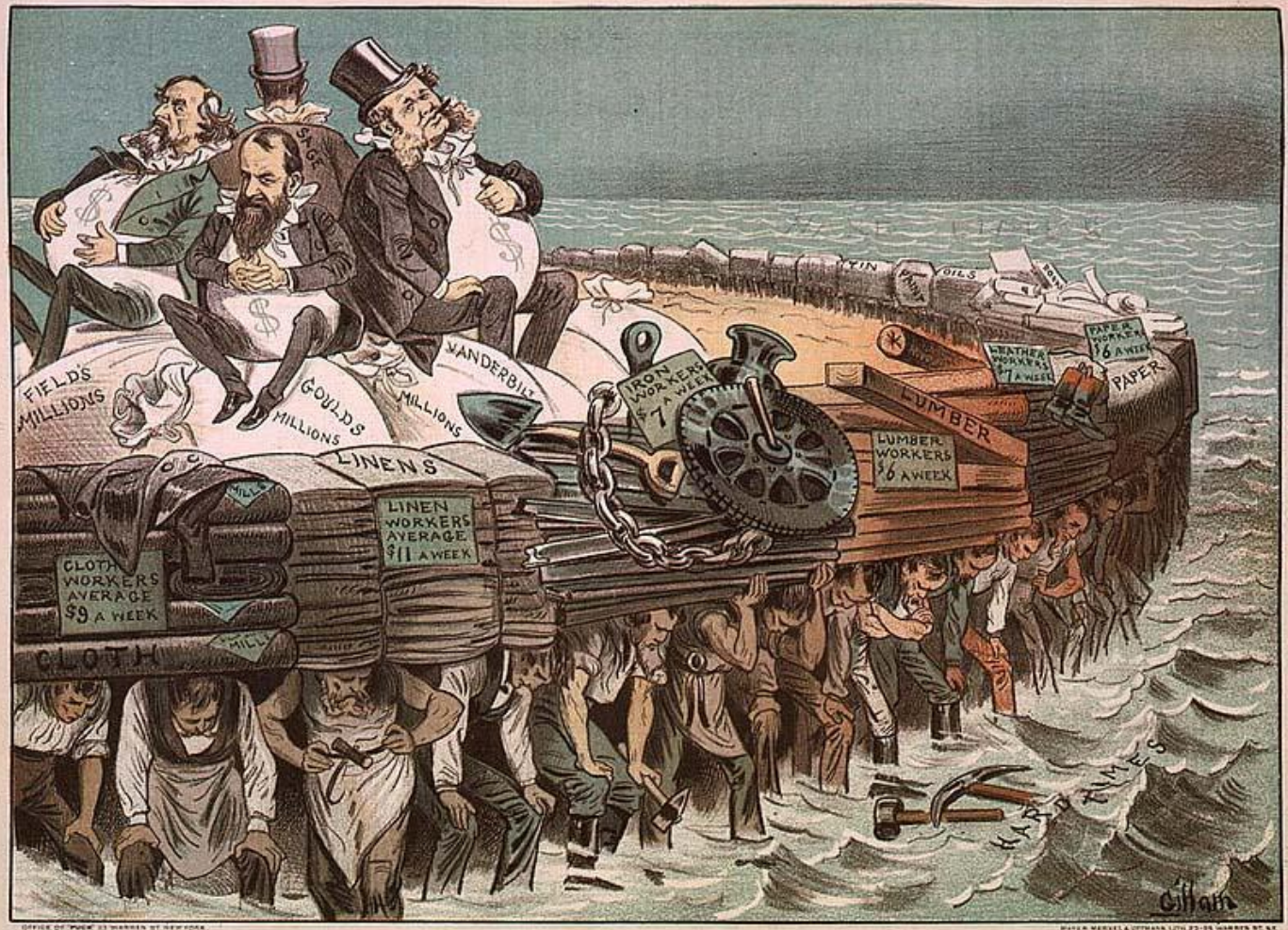
Explain and analyze the ways in which different groups of society responded to the impacts of the Industrial Age

Reevaluating Life and Society



Changing Cities - Urban Renewal, Skyscrapers, and Improved Standards of Living

Corporations and Monopolies



THE PROTECTORS OF OUR INDUSTRIES.

Changes to Societal Structures

- New Class Structures
 - Upper Class, Upper Middle, Middle, Lower Middle, Working Class, Poor
- Cult of Domesticity
 - Women stay home
 - Children have strict guidelines to follow based on gender



The Push for Women's Rights

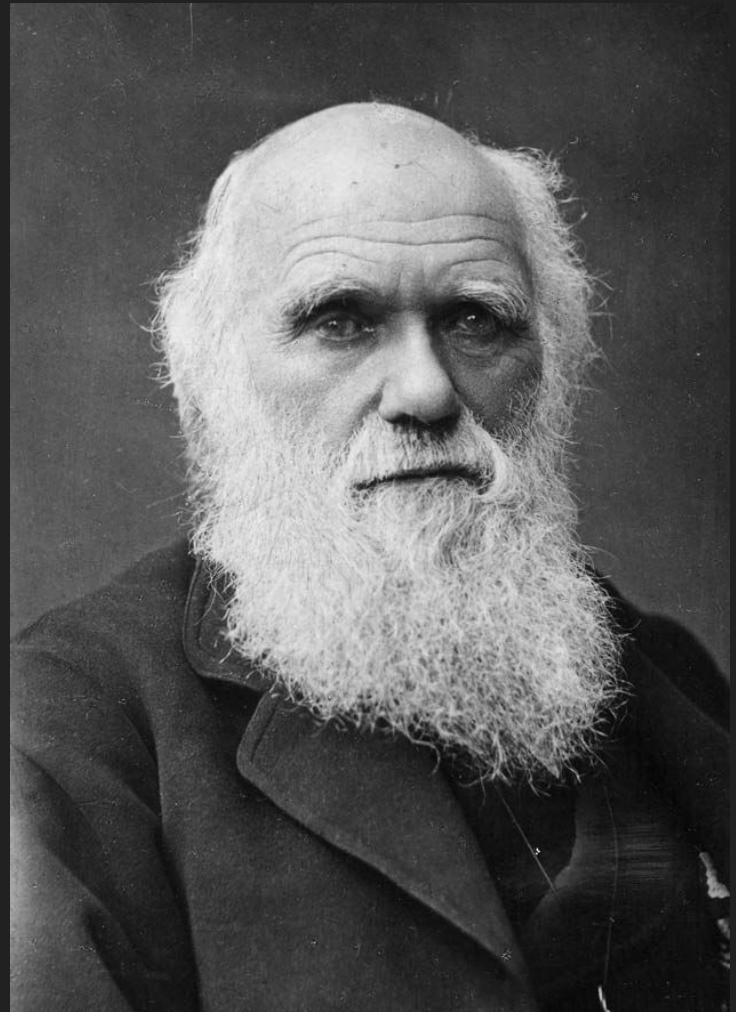
- The Temperance Movement
 - Limit or ban the use of alcoholic beverages
- Suffrage Movement
 - Combination of women from all different time periods and walks of life
 - Lucretia Mott, Elizabeth Cady Stanton, Susan B. Anthony, Sojourner Truth, Alice Paul



CICELY IN THE SALOON.

New Ways of Thinking

- Charles Darwin, *On the Origin of Species* (1859)
 - Natural Selection and Social Darwinism
 - Scientific Racism
- Religious Backlash in the form of the Social Gospel



Social Darwinism Document Questions

1. In one sentence, summarize the author's argument.
2. Provide three arguments in favor of the author's ideas.
3. Provide three arguments against the author's ideas.
4. What do you believe about the topic?

New Ways and Forms of Expression



- Romanticism
 - Emphasizes Imagination, Freedom, and Emotions
- Realism
 - Represent the world as it was and is without exaggeration
- Impressionism and Postimpressionism

Romanticism Vs. Realism

Romanticism

- Emphasizes imagination and emotions
- Depicts larger-than-life heroes
- Views the world idealistically
- Focuses on the exotic, the supernatural and imaginary worlds



Realism

- Emphasizes accuracy and objectivity
- Depicts common, everyday heroes
- Views the world scientifically
- Focuses on real situations



IMPRESSIONISM

VERSUS

POST IMPRESSIONISM

IMPRESSIONISM

An art movement that originated in the late 19th century in Paris as an artistic reaction to the rapidly changing urban environment

Characterized by the use of vibrant colors, harsh spontaneous brushstrokes, accurate depiction of light with its changing quality, use of unusual visual angles, and the use of subjects from the modernized urban life

POST IMPRESSIONISM

An art movement that originated in the same century in France as a reaction to impressionism

Diverse characteristics and styles that differ from artist to artist as there were many individual artist painting styles such as Pointillism, Japonisme related painting style, pictorial form style, Synthetism style, Primitivism style, etc.

CURIOUS
MUSE

An Impressionist painting of two women. The woman on the left wears a wide-brimmed straw hat and a red high-collared dress. The woman on the right wears a dark, ornate hat and a dark coat with a high collar. The background is a vibrant red with a large white abstract shape on the left and colorful geometric confetti (squares, circles, triangles) scattered throughout. The text 'CURIOUS MUSE' is in the top left, and 'IMPRESSIONISM EXPLAINED' is at the bottom in large white letters.

IMPRESSIONISM

EXPLAINED

CURIOUS
MUSE

**POST-
IMPRESSIONISM**
EXPLAINED

Romanticism

- https://docs.google.com/document/d/1kSosNABkgLIQfHWdEpfvRUmSR96NN4AHxm_HvrQ5JMA/edit?usp=sharing
- <https://www.youtube.com/watch?v=fQk8Tm815IE>
 - Beethoven's 5th Symphony
- <https://www.youtube.com/watch?v=fRg0K5rgXog>
 - Beethoven's 6th Symphony
- <https://www.youtube.com/watch?v=WgrusoQ6xVM>
 - Franz Liszt

Realism

- https://docs.google.com/document/d/1kSosNABkgLIQfHWdEpfvRUmSR96NN4AHxm_HvrQ5JMA/edit?usp=sharing
- [Gustave Courbet - The Stonebreakers - WGA05457.jpg](#)
- [GrossClinic1.jpg](#)

Impressionism and Postimpressionism

- Impressionism
 - [Impressionism_2.jpg](#)
 - [Impressionism_6.jpg](#)
 - [DG1118-1000x1000.jpg](#)
 - [Claude-Monet-Waterlilies-and-Japanese-Bridge-1899-HIGH-RES.jpg](#)
 - [jeune-fille-dans-le-jardin-de-giverny-claude-monet.jpg](#)
- Postimpressionism
 - [Starry_night_full.jpg](#)
 - [Post-impressionism-16.jpg](#)
 - [Post-impressionism-17.jpg](#)
 - [post-impressionism-19.jpg](#)

Lesson 4 Objectives

Identify ways in which the Industrial Age reshaped the social and economic structures of cities

Explain and analyze the ways in which different groups of society responded to the impacts of the Industrial Age